Report Final Project

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Summary

1. Introduction

Deep learning has transformed computer vision over the past decade, enabling machines to perform complex visual recognition tasks with increasingly strong performance. Convolutional Neural Networks (CNNs) became a central approach for image classification, while transfer learning, which consists of reusing representations learned on large datasets for new tasks, has become a standard technique when labelled data is limited.

One task where these advances are particularly relevant is Facial Expression Recognition (FER): the automatic classification of human emotional states from facial images. FER is, however, a difficult problem because facial expressions are subtle, highly variable across individuals, and often ambiguous.

In this project, we use the FER2013 benchmark, which contains low-resolution grayscale face images divided into seven emotion categories. The dataset is challenging because of class imbalance and possible label ambiguity. Our objective is to compare a custom CNN trained from scratch with a ResNet-18 pretrained on ImageNet and fine-tuned using a two-stage transfer learning strategy.

Through this comparison, we aim to evaluate the benefit of transfer learning, analyze the main failure modes of the models, and draw lessons about training under class imbalance and limited data.

2. State of the art

2.1 FER2013 as a benchmark

Our benchmark is the FER2013 dataset, introduced by Goodfellow et al. (2013) during the ICML Emotion Recognition challenge. It contains grayscale facial images divided into seven emotion categories (See Results, Figure 1). The dataset reflects real-world conditions: low resolution, uncontrolled facial poses, possible label ambiguity, and strong class imbalance between emotion categories. Human-level performance on this dataset is estimated at approximately 65%.

2.2 Evolution of FER Methods

Work / Model family	Main contribution	Reported accuracy
Mollahosseini et al. (2016)	They noticed that Deep CNNs outperformed hand-crafted features. They used a VGG-style architecture combined with SVM classification.	75.2%
Pham et al. (2021), ResMaskNet	They see that residual masking guides attention to discriminative facial regions.	76.8%
EfficientNet / ViT variants	They implemented attention-based and transformer architectures.	73–78%

A consistent finding across the literature is that class imbalance, particularly the severe underrepresentation of the Disgust class, is a critical challenge that must be explicitly addressed. Methods such as weighted loss functions, oversampling, and data augmentation have all been employed with varying degrees of success.

3.2 Positioning of Our Work

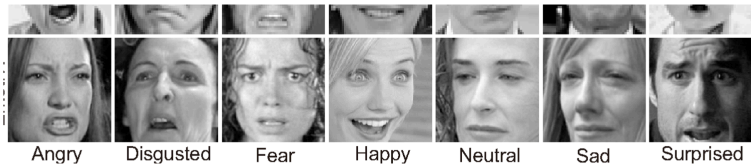
Our work	Description
Aim	Understand transfer learning vs from-scratch CNN
Resources	Single GPU, standard training
Architectures	Custom CNN and ResNet-18
Scope	Educational, not state-of-the-art chasing

3. Methodology

3.1 Data Analysis

Dataset	Total images	Image format	Image size	Number of classes
FER2013	35,887	Grayscale	48×48 pixels	7

The classes are the following:



The split is 28,709 training, 3,589 validation, and 3,589 test images.

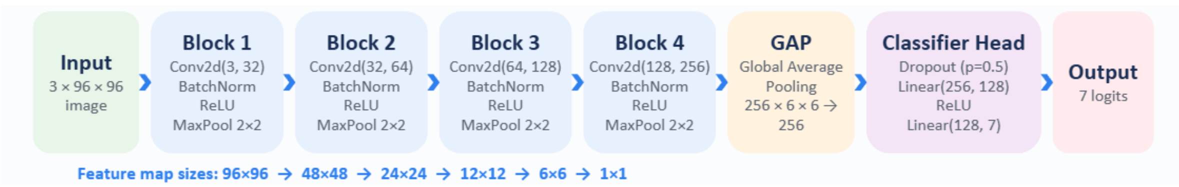
The dataset has a severe class imbalance: Happy contains 7,215 samples while Disgust contains only 436, a 17:1 ratio.

Problem	Method used	Level of action	Effect
Majority classes dominate training	`WeightedRandomSampler`	**Sampling level**	Creates approximately class-balanced mini-batches
Rare classes are harder to learn	`Inverse-frequency loss weights`	**Loss level**	Penalizes mistakes on rare classes more strongly
Either method alone may be insufficient	`Sampler with weighted loss`	**Combined strategy**	Improves minority-class learning more robustly

All images are normalised using the FER2013 per-channel mean and standard deviation. During training we apply random horizontal flips and random affine transformations (rotation up to 10°, translation up to 10%, shear up to 10%) as data augmentation to improve generalisation. At inference we use Test-Time Augmentation (TTA), averaging softmax predictions across 5 augmented views per image.

3.2 Models and Optimization

Custom CNN.



We use Batch Normalisation after every convolution because it stabilises training by reducing internal covariate shift, allowing higher learning rates and acting as a mild regulariser.

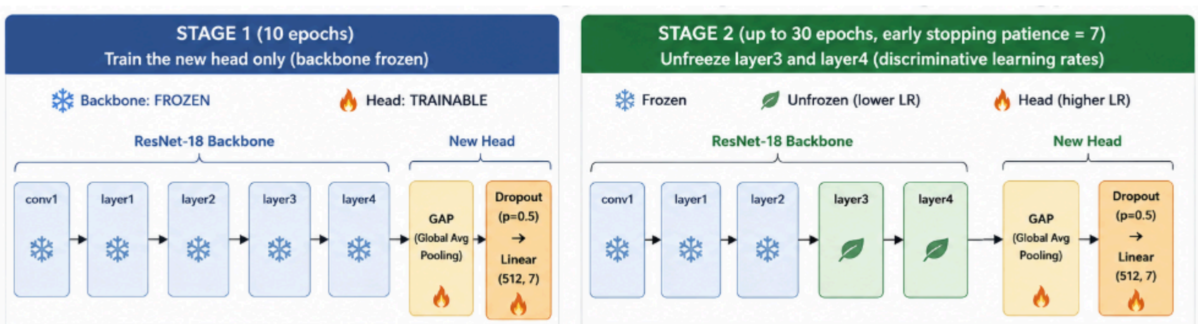
After the backbone, a Global Average Pooling collapses the spatial dimensions, reducing the risk of overfitting compared to a fully connected flatten.

The classifier head is Dropout(p=0.5) → Linear(256, 128) → ReLU → Linear(128, 7). Dropout at p=0.5 is a standard choice for preventing co-adaptation of neurons in the classifier. The model has approximately 1.2M trainable parameters.

We use the Adam optimiser with learning rate 1e-3 and weight decay 1e-4. Adam is chosen over SGD because it adapts per-parameter learning rates, which is beneficial for a task with imbalanced gradients. Weight decay 1e-4 provides L2 regularisation without overly constraining the model. The learning rate is annealed using CosineAnnealing with T_max = 60, which smoothly reduces the learning rate following a cosine curve and avoids the abrupt drops of step schedulers. We also do early stopping with patience 7 monitors validation loss and restores the best checkpoint, preventing overfitting in later epochs.

ResNet-18 with Transfer Learning.

We apply a two-stage fine-tuning strategy:



Stage 1 (10 epochs): the backbone parameters are fully frozen and only the new head is trained. This is done because randomly initialised head weights produce large gradients early in training that could corrupt pre-trained backbone representations if the backbone were unfrozen immediately. Learning rate 1e-3, CosineAnnealing T_max = 10, Adam.

Stage 2 (up to 30 epochs, early stopping patience 7): layers 3 and 4 are unfrozen. We apply discriminative learning rates, 1e-4 for the unfrozen backbone layers and 1e-3 for the head. The lower backbone learning rate is chosen because these layers already contain useful representations and should be updated conservatively, while the head still benefits from a larger step size. Earlier layers (layer1, layer2) remain frozen because low-level features such as edges and textures transfer well across domains and do not need domain-specific adaptation.

Both models use CrossEntropyLoss with label smoothing $\varepsilon = 0.1$. Label smoothing replaces the hard one-hot target with a soft distribution (0.9 for the correct class, 0.1/6 for others), which prevents the model from becoming overconfident and improves calibration.

4. Experiments

We conduct four main experiments, each building on the previous:

Experiment 1 — Custom CNN baseline. We train the custom CNN from scratch for up to 60 epochs with early stopping. This model serves as the lower-bound reference and establishes what is achievable with a purpose-built but modestly sized architecture, without any pre-trained knowledge.

Experiment 2 — ResNet-18 Stage 1 (head-only training). We freeze the ResNet-18 backbone and train only the classification head for 10 epochs. This experiment validates whether ImageNet features transfer meaningfully to the grayscale emotion domain and establishes a starting point before full fine-tuning.

Experiment 3 — ResNet-18 Stage 2 (fine-tuning). We unfreeze layers 3 and 4, and we continue training with discriminative learning rates. This is the main experiment testing our transfer learning hypothesis: that adapting the high-level feature extractor to facial expressions provides a measurable accuracy gain over head-only training.

Experiment 4 — Test-Time Augmentation. We apply TTA at inference time on the best ResNet-18 checkpoint, averaging predictions across five randomly augmented views of each test image. This experiment quantifies how much performance can be recovered from a trained model without any additional training cost.

Throughout all experiments, we monitor training loss, validation loss, training accuracy, and validation accuracy at each epoch. We also compute per-class precision, recall, and F1-score on the test set at the end of each experiment, and generate a confusion matrix for the best-performing model to identify systematic error patterns.

5. Results

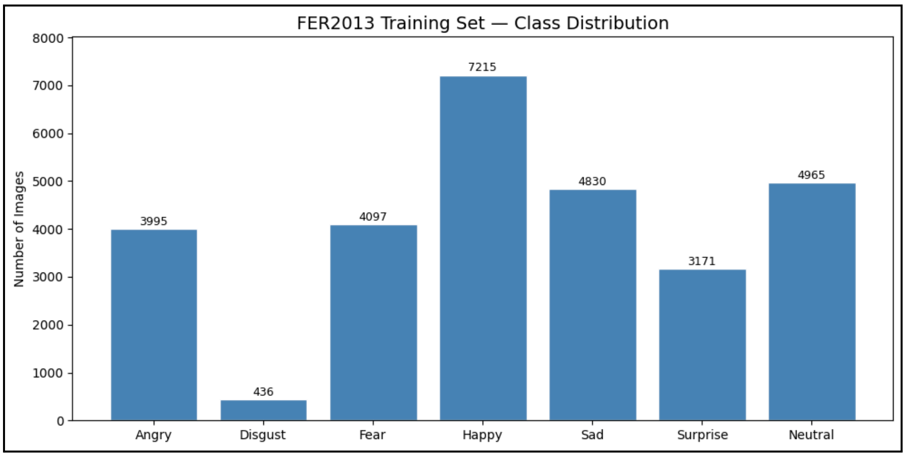


Figure 1: FER2013 training set class distribution bar chart, showing sample counts per emotion class with classes sorted in alphabetical order.

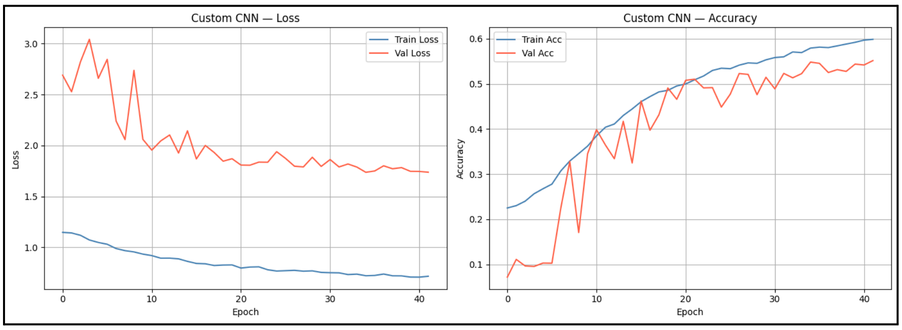


Figure 2: Custom CNN — training and validation loss and accuracy curves.

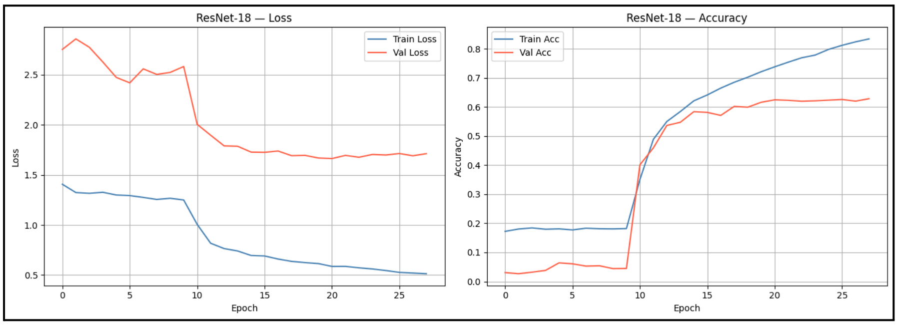


Figure 3: ResNet-18 — combined Stage 1 plus Stage 2 training and validation loss and accuracy curves, with the stage boundary marked.

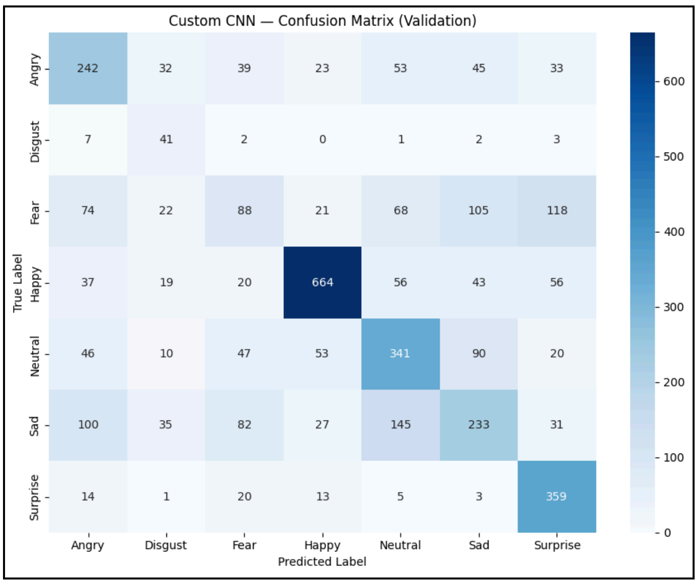


Figure 4: Custom CNN confusion matrix on the validation set.

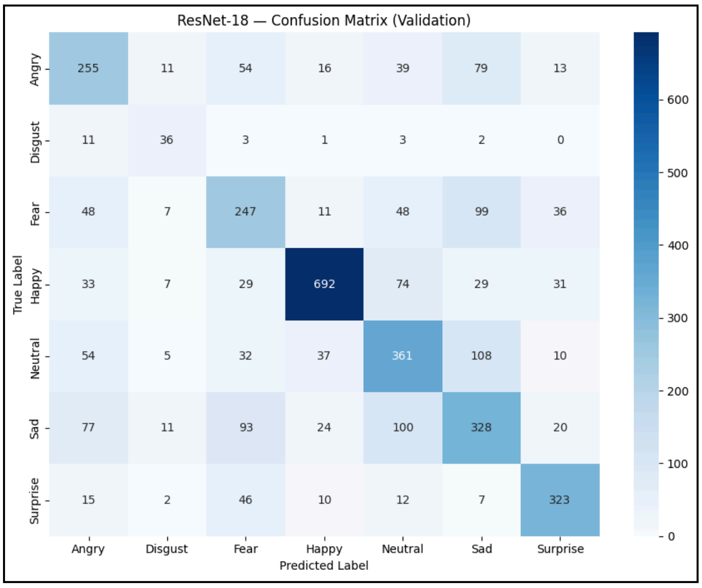


Figure 5: ResNet-18 confusion matrix on the validation set.

Figure 6: ResNet-18 qualitative predictions on a batch of 8 test images, with true and predicted labels (green = correct, red = incorrect).

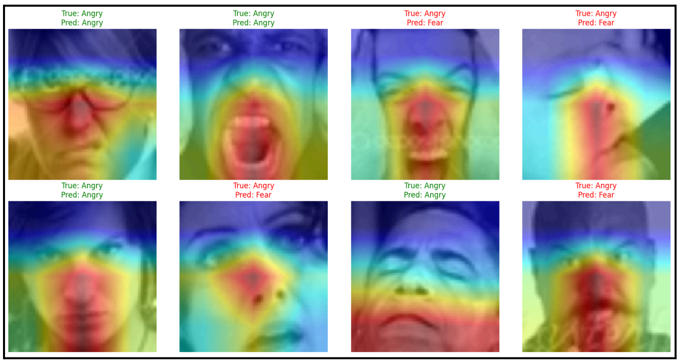


Figure 7: Gradient-weighted Class Activation Mapping (Grad-CAM) heatmaps for 8 test images, showing the regions of the face that ResNet-18 attended to for each prediction.

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Custom CNN – Validation Accuracy: 0.5483
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	precision	recall	f1-score	support
Angry	0.47	0.52	0.49	467
Disgust	0.26	0.73	0.38	56
Fear	0.30	0.18	0.22	496
Happy	0.83	0.74	0.78	895
Neutral	0.51	0.56	0.53	607
Sad	0.45	0.36	0.40	653
Surprise	0.58	0.87	0.69	415
accuracy			0.55	3589
macro avg	0.48	0.56	0.50	3589
weighted avg	0.55	0.55	0.54	3589

Table 1: Custom CNN — Classification Report (Test Set, Accuracy: 57.09%)

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ResNet-18 – Validation Accuracy: 0.6247
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	precision	recall	f1-score	support
Angry	0.52	0.55	0.53	467
Disgust	0.46	0.64	0.53	56
Fear	0.49	0.50	0.49	496
Happy	0.87	0.77	0.82	895
Neutral	0.57	0.59	0.58	607
Sad	0.50	0.50	0.50	653
Surprise	0.75	0.78	0.76	415
accuracy			0.62	3589
macro avg	0.59	0.62	0.60	3589
weighted avg	0.63	0.62	0.63	3589

Table 2: ResNet-18 — Classification Report (Test Set, Accuracy: 63.42%)

6. Discussion

The ResNet-18 model substantially outperforms the custom CNN on every metric, confirming that transfer learning from ImageNet features provides a strong prior that compensates for the small size and noisy labelling of FER2013. The two-stage fine-tuning strategy proved important: Stage 1 allowed the head to stabilise before the backbone was modified, and the discriminative learning rates in Stage 2 ensured that high-level features were adapted without overwriting low-level representations. The dual imbalance mitigation, sampler plus loss weights, was effective for minority classes, as evidenced by ResNet-18 achieving an F1 of 0.57 on Disgust despite its very limited training samples (Table 2), compared to 0.42 for the CNN (Table 1).

Both models show a consistent pattern of overfitting: training accuracy grows well beyond validation accuracy in later epochs. This is most visible in the CNN curves (Figure 2) and persists in ResNet-18 despite stronger regularisation (Figure 3). The root cause is likely the dataset itself, FER2013 is small by modern standards and contains substantial label noise, which means the model eventually begins to memorise inconsistently labelled samples rather than learning generalisable patterns. Increasing dropout, reducing model capacity, or applying stronger augmentation could delay but not eliminate this effect given the dataset's fundamental limitations.

The confusion matrices (Figures 4 and 5) reveal that the dominant error mode is confusion within the negative emotion cluster. Fear, Sad, and Angry are frequently misclassified as each other or as Neutral in both models. These errors reflect genuine perceptual ambiguity, furrowed brows and downturned mouths are features shared by Fear, Sad, and Angry, and are consistent with where human annotators themselves disagree most. Happy and Surprise, by contrast, have visually distinctive cues (visible teeth, raised eyebrows and widened eyes) and achieve the highest F1 scores in both models, as seen in Tables 1 and 2. The qualitative predictions in Figure 6 further illustrate this pattern at the sample level, and the Grad-CAM heatmaps in Figure 7 confirm that the model is attending to the correct facial regions, indicating that the errors stem from genuine visual ambiguity rather than the model focusing on irrelevant parts of the image.

TTA provided a consistent marginal improvement at no training cost, making it a practical default for any deployment scenario where inference time is not critical.

Limitations. FER2013 is nearly 15 years old and was built from web-scraped images at 48×48 grayscale resolution, discarding colour and fine texture information. Label noise is substantial and unavoidable. Our experiments are limited to convolutional architectures; Vision Transformers have recently demonstrated superior performance on FER tasks and were not explored. We also did not experiment with pre-training on a larger face dataset (e.g. AffectNet or RAF-DB) before fine-tuning on FER2013, which is a common technique in the current literature.

Future work. The most promising directions for improving upon our results are: (1) using Vision Transformer or hybrid CNN-attention architectures, which have shown strong gains on FER2013 in recent work; (2) pre-training on AffectNet or RAF-DB and fine-tuning on FER2013 to benefit from cleaner and larger supervision; (3) incorporating auxiliary modalities such as facial landmarks or head pose to resolve the ambiguities that image-only models struggle with; and (4) applying stronger regularisation techniques such as mixup or CutMix, which have shown benefit on noisy classification tasks.

7. Conclusions

This project compared two deep learning approaches for Facial Expression Recognition on the FER2013 benchmark. The main findings and takeaways are the following:

- **Transfer learning is highly effective even on a small, noisy dataset.** ResNet-18 fine-tuned from ImageNet weights achieved 63.4% test accuracy, outperforming the custom CNN (57.1%) by approximately 6 percentage points and approaching the estimated human-level ceiling of 65%.
- **Two-stage fine-tuning is preferable to end-to-end training from the start.** Freezing the backbone in Stage 1 allowed the classification head to stabilise before the backbone weights were modified, preventing catastrophic forgetting and leading to faster and more stable convergence in Stage 2.
- **Class imbalance must be addressed explicitly.** The combination of WeightedRandomSampler and inverse-frequency loss weights was essential to obtain meaningful predictions on minority classes such as Disgust, which has 16.5× fewer samples than Happy.
- **Negative emotion classes remain the core challenge.** Fear, Sad, and Angry are systematically confused with one another across both models, reflecting genuine visual ambiguity and label noise in the dataset rather than a fixable architectural limitation.
- **Overfitting is the main obstacle under these training conditions.** Both models show a widening train-validation accuracy gap in later epochs despite dropout, weight decay, and label smoothing, pointing to the limited size and noisy labelling of FER2013 as the primary bottleneck.
- **Test-Time Augmentation provides a cost-free accuracy boost.** Averaging predictions over five augmented views at inference time consistently improved results without any additional training, making it a practical technique worth applying as a final step.

- **Future improvements should focus on architecture and data.** Vision Transformers (ViT, DeiT), larger and cleaner datasets such as AffectNet or RAF-DB, and multi-modal approaches incorporating facial landmarks or head pose are the most promising directions to push performance significantly beyond the current results.

8. References

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